

Smart Home Tips: Avoiding water damage from plumbing leaks

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Potential for damage

Outer Banks property owners are all keenly aware of the risks of flooding and storm damage posed by living on a barrier island. We do what we can to protect ourselves and our property to mitigate those risks.

Our cottages face another persistent risk of water damage — the threat of leaks, large and small, from plumbing and appliances. A broken ice maker, ruptured washing machine hose, cracked toilet bowl, or a failed pipe can bring devastating damage to the interior of a home if not addressed quickly. While any home can have these issues, it is especially damaging when the home is not occupied throughout the year with no one checking on it daily.

Luckily, new technology exists to not only alert residents and absentee homeowners, but to proactively stop the flow of the damaging water as soon as it is detected.

Consider these statistics

According to the Insurance Information Institute[1], one in 60 insured homes had a claim due to water damage or freezing. While not distinguishing between storm-related and internal leaks, The Hartford reports[2] that water and freezing damage amounted to 20% of the insurance claims in 2025 with an average cost of \$35,000.

Water waste through leaks

The EPA reports[3] that homes with undetected water leaks average 9,300 gallons of wasted water per year, and that 9% of all homes have leaks that waste 50 gallons or more per day. These leaks may not do damage if they flow into the home's wastewater system but it costs the homeowners money in increased water bills, wastes our community's precious freshwater resources, and puts extra demand on municipal sewer systems, or worse, septic tanks.

Technical solutions

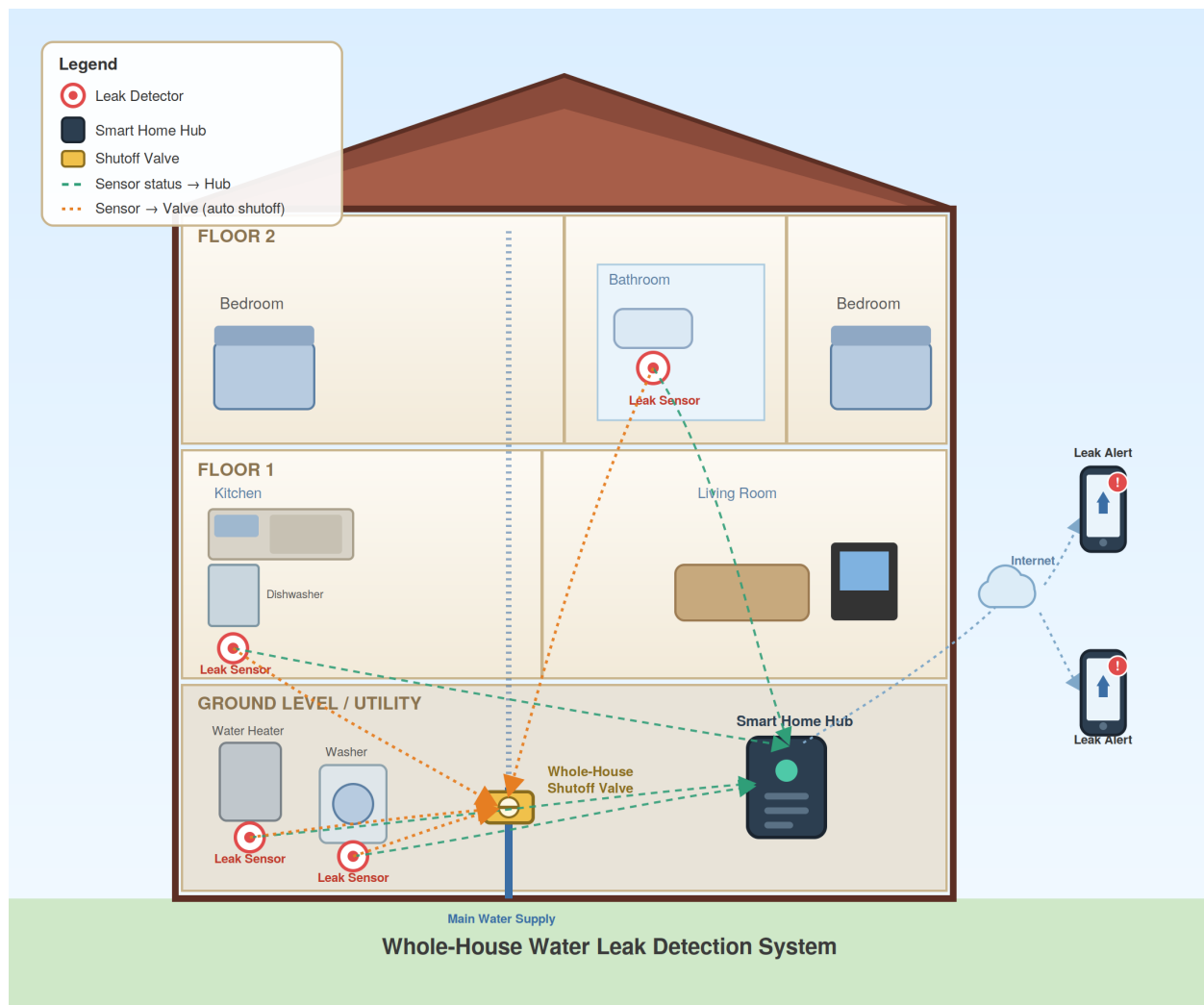
While the statistics show the risk of leaks from plumbing and appliances is substantial, there are now tools that can help reduce the risk, prevent the damage, and give the homeowner real peace of mind.

This article will discuss three tools that can be used together in a home to achieve the goal of detecting water leaks and taking action, often with no human intervention required.

Overview

The entire system consists of four different parts: a hub, leak detectors, a water flow monitor (optional), and an automated shut-off valve.

The diagram shows a typical configuration with four leak detectors, a hub, and a smart shut-off valve. Each of the components is described below.



Leak detectors

The first tool is a simple leak detector, which is a small (2 3/8" x 2 3/8" x 1"), inexpensive, battery-powered device which can be placed around the home in locations prone to leaking. These locations would include sinks, toilets, dishwashers, washing machines, refrigerators, bathtubs, and water heaters.

The detectors have sensors on the top and bottom of the unit so they can detect water from above or pooling on the floor. When detected, a loud alarm sounds and a signal is sent to the hub and the shut-off valve over a long-range protocol called LoRa[4]. For the automated system to work, no Wi-Fi or internet connection is required.

A five-bedroom home with five bathrooms would need about a dozen detectors throughout the home. Each costs about \$16 for a total of less than \$200.

Whole-house automated shut-off valves

The second component is a smart shut-off valve to turn off all water coming into the house when a leak is detected. The valve will shut automatically when one of the leak detectors senses water and it can also be controlled via a smartphone app.

Water flow monitors

While most water leaks will happen at the fixtures and appliances where a leak detector is placed, other leaks can happen in the piping in the wall or by flow that does not escape onto the floor such as a stuck toilet flapper. A water flow monitor can detect these unusual flow patterns and either cause the valve to close or alert the homeowner via a notification. The monitor can be a part of the shut-off valve or a separate unit. The monitor clamps onto the incoming water pipe and uses ultrasonic sensors to measure the water flow. This is the same technology used in some public water meters, such as the ones recently installed by Nags Head.

Installation

All components described can be installed by a homeowner except for the shut-off valve, which should be installed by a qualified plumber.

Contact Brad at Smart Homes OBX if you have questions or would like help designing a system for your home.

References

[1] <https://www.iii.org/fact-statistic/facts-statistics-homeowners-and-renters-insurance>, accessed 2026-08-24.

[2] <https://www.businesswire.com/news/home/20251209433403/en/Water-And-Freezing-Damage-Burglary-Lead-The-Hartfords-Top-Five-Small-Business-Claims>, accessed 2026-08-25

[3] <https://www.epa.gov/watersense/fix-leak-week>, accessed 2026-08-24.

[4] <https://en.wikipedia.org/wiki/LoRa>

Further reading

<https://www.epa.gov/watersense/leak-detection-and-flow-monitoring-devices>